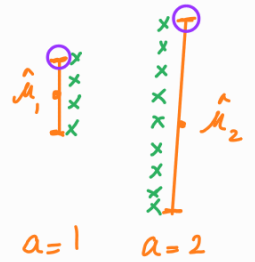


Online Learning over MDP

Bandit algorithms have inspired online learning algorithms for general reinforcement learning over MDPs. A promising line of work relies on the principle of optimism in the face of uncertainty.



offline learning: suboptimality gap = $V^*(\mathcal{M}_0) - V^{\pi_t}(\mathcal{M}_0) \leq \epsilon$ w.p. $1 - \delta$

online learning: cumulative regret = $\sum_{t=1}^T V^*(\mathcal{M}_0) - V^{\pi_t}(\mathcal{M}_0) \leq B$ w.p. $1 - \delta$

Upper Confidence Reinforcement Learning: In the online learning setting with unknown model, the UCR1 algorithm

- model-based • maintains confidence bounds on the unknown transition function and reward functions, and
- computes optimistic policies by choosing optimistic models that lead to the highest average reward.

Interaction: The agent interacts with the environment in phases such that certain number of visits to a state-action pair are obtained.

Counting statistics: Using the history (information) up to time t , we compute the following counting statistics

$$N_t(s, a, s') = \sum_{t'=1}^{t-1} \mathbb{1}[(s_{t'}, a_{t'}, s_{t'+1}) = (s, a, s')]$$

$$N_t(s, a) = \sum_{t'=1}^{t-1} \mathbb{1}[(s_{t'}, a_{t'}) = (s, a)]$$

Transition Estimation: The transition probabilities are estimated by

$$\hat{P}_t(s' | s, a) = \frac{N_t(s, a, s')}{\max\{1, N_t(s, a)\}} \quad \forall s \in S, a \in A, s' \in S$$

Reward Estimation: The reward is estimated by

$$\hat{R}_t(s, a) = \frac{1}{\max\{1, N_t(s, a)\}} \sum_{t'=1}^{t-1} \mathbb{1}[(s_{t'}, a_{t'}) = (s, a)] r_{t'}$$

Building Confidence sets: The set of all MDPs with transition

probabilities close to $\hat{P}_t$ and rewards close to $\hat{R}_t$

is denoted by the confidence set $C_t(\mathcal{M})$ defined

by

confidence set for MDP $\leftarrow C_t(\mathcal{M}) = \{ \mathcal{M} \text{ such that}$

$$\| P(\cdot | s, a) - \hat{P}_t(\cdot | s, a) \|_1 \leq c_1 \sqrt{\frac{|S| \log(|A| t / \delta)}{\max\{1, N_t(s, a)\}}}$$



confidence set for P



confidence set for R

$\forall s \in S, \forall a \in A$, and

$$|R(s, a) - \hat{R}_t(s, a)| \leq c_2 \sqrt{\frac{\log(|S||A| t / \delta)}{\max\{1, N_t(s, a)\}}} \quad \forall s \in S, \forall a \in A \quad \}$$

Extended value Iteration: The extended value iteration yields an optimistic value function and policy.

- Initialize $V_0^t(s) = 0$ → time / round
- For $k = 0, 1, 2, \dots$ → iteration

$$V_{k+1}^t(s) = \max_{a \in A} \left[\max_{\tilde{R}_t} \tilde{R}_t(s, a) + \max_{\tilde{P}_t} \mathbb{E}_{s' \sim \tilde{P}_t(\cdot | s, a)} [V_k^t(s')] \right] \quad \forall s \in S$$

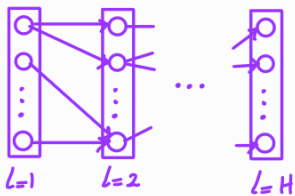
Theorem: In the online learning setting over MDPs with unknown model, the UCRL2 algorithm satisfies the following regret upper bound with probability at least $1 - \delta$:

$$\text{Reg}(T) = \mathbb{E} \left[\sum_{t=1}^T (V^*(s_0) - V^{\pi_t}(s_0)) \right] \leq O(D|S| \sqrt{|A| T \log(T/\delta)})$$

Upper Confidence Bound Value Iteration: In the online learning setting without oracle access, the UCB-VI algorithm

- model-based**
- maintains estimates of the unknown transition function and reward functions, and
 - computes optimistic policies by performing value iteration on the estimated transition and the estimated reward augmented with an exploration bonus.

Interaction: The agent interacts with the environment in episodes of length H , resulting in trajectories



$$\tau^i = (s_0^i, a_0^i, r_0^i, s_1^i, a_1^i, r_1^i, \dots, s_{H-1}^i, a_{H-1}^i, r_{H-1}^i)$$

for $i = 1, 2, \dots, T$.

Counting statistics: Using the history (information) up to time t , we compute the following counting statistics

$$N_t(s, a, s') = \sum_{t'=1}^{t-1} \sum_{h=0}^{H-1} \mathbb{1}[(s_h^{t'}, a_h^{t'}, s_{h+1}^{t'}) = (s, a, s')]]$$

$$N_t(s, a) = \sum_{t'=1}^{t-1} \sum_{h=0}^{H-1} \mathbb{1}[(s_h^{t'}, a_h^{t'}) = (s, a)]$$

Transition Estimation: The transition probabilities are estimated by

$$\hat{p}_t(s' | s, a) = \frac{N_t(s, a, s')}{N_t(s, a)} \quad \forall s \in \mathcal{S}, a \in \mathcal{A}, s' \in \mathcal{S}$$

Reward Estimation: The reward is estimated by

$$\hat{R}_t(s, a) = \frac{1}{N_t(s, a)} \sum_{t'=1}^{t-1} \sum_{h=0}^{H-1} \mathbb{1}[(s_h^{t'}, a_h^{t'}) = (s, a)] r_h^{t'}$$

Exploration Bonus: The exploration bonus added to the reward is defined as

$$b_t(s, a) = 2H \sqrt{\frac{L}{N_t(s, a)}} \quad \text{where } L = \log\left(\frac{|\mathcal{S}||\mathcal{A}|HT}{\delta}\right)$$

Optimistic Value Iteration: Using the augmented reward, the value iteration can be done optimistically

- Initialize $\hat{v}_H^t(s) = 0 \quad \forall s \in \mathcal{S}$

- For $h = H-1, H-2, \dots, 0$

- $$\hat{Q}_h^t(s, a) = \hat{R}_t(s, a) + b_t(s, a) + \mathbb{E}_{s' \sim \hat{p}_t(s, a)} [\hat{v}_{h+1}^t(s')] \quad \forall s \in \mathcal{S}, a \in \mathcal{A}$$

- $$\hat{v}_h^t(s) = \max_{a \in \mathcal{A}} \hat{Q}_h^t(s, a) \quad \forall s \in \mathcal{S}$$

UCB-VI Algorithm

- Initialize $\hat{P}_1$, $\hat{R}_1 = 0$, and b_1
- For $t = 1, 2, \dots, T$:
 - Compute an optimistic policy π_t by value iteration using $\hat{P}_t$ and $\hat{R}_t + b_t$
 - Generate (sample) a trajectory according to π_t
 - Update the estimated transition function $\hat{P}_{t+1}$
 - Update the estimated reward function $\hat{R}_{t+1}$
 - Update the exploration bonus b_{t+1}

Theorem: In the online learning setting with finite horizon over MDPs with unknown model, the UCB-VI algorithm satisfies the following regret upper bound: → with $\delta = \frac{1}{TH}$

$$\begin{aligned} \text{Reg}(T) &= \mathbb{E} \left[\sum_{t=1}^T (v^*(s_0) - v^{\pi_t}(s_0)) \right] \\ &\leq O(H^2 |S| \sqrt{|A| T \log(|S| |A| H^2 T^2)}) \end{aligned}$$